Meru Cabs embraces open source technologies day in and day out, and is proud to do that as well. Sangoi asserts, "Today, I don't think IT decision-makers can say that they don't use open source technology. Open source is a part of life for people like us nowadays. Even the traditional vendors, who operated on an entirely different business model compared to open source, have started embracing open source technology. Open source is becoming mainstream and will only get bigger and better."

"Our Web services back-end, our mobile app back-end, et al, is on open source. Whether it is about network management, network monitoring or analytics, we use various open source tools on a regular basis. We use the LAMP stack most frequently. This is where our entire mobile and Web architecture backend lies. We use Redis, which is an in-memory data store. For analytics, most of our work is done in R. And we use MySQL at the database level," Sangoi adds.

Meru Cabs has an engineering team of about 100 people. About 50 per cent of the team members are very adept at open source technologies, shares Sangoi.

The many advantages

Sangoi says, "Open source technologies allow you to tweak products much more than any proprietary product, as the source code is available right up to the product level. The performance of an open source product is better, because a lot of people are continuously contributing to it and hence the product is constantly evolving. In many cases, the issues that you may encounter will have already been encountered and resolved by someone else. Community support is very strong, which is very advantageous. There is also a definite advantage with regard to price as well. At times, when you don't want to spend on paid support, you can get options at a lower cost, or even zero cost. For example, if you want to put a Linux server for a mission critical job, you might want to go for a Red Hat subscription. But if you are using a tool like R for any of your back-end processes, you may not require support. So such decisions are taken based on the requirements. It is about taking calculated risks when it comes to deploying open source technologies."

What matters is the solution

According to those in the company, the decision to adopt open source technology was not based on an ideological choice between open source or closed source software, but was about the best possible solution. Sangoi shares, "For any task, we first look at the options that fit best. We don't claim that we are 100 per cent open source, but based on the specific demands for a particular task or technology, we look at what are the best options available, proprietary or open source solutions, to solve that business problem. There are definitely times when open source comes out as the best option. More and more, we are moving towards open source technologies. We evaluate open source technologies and the support available. If we are satisfied and if our initial trials are successful, then we look at adopting those open source tools."

Adopting open source becomes easier if an IT decision-maker gets the support of the management. Nilesh Sangoi, fortunately, gets a lot of it. He explains, "Yes, people do have their inhibitions about using open source technologies, and there are some questions that keep coming our way. Once we explain those aspects to the management, and show case studies where these technologies are being used in other large organisations, or start-ups for that matter, then people are fairly open to using these technologies. On some occasions, we prefer showcasing the actual product, which is a proof-of-concept, to convince people that open source works well."

The road to open source can be bumpy

While Sangoi is happy adopting open source tools and technologies, he has faced his share of issues too. He says, "While there is a lot of support available, yet a lot of it is still about 'doing it yourself'. In many cases, the user needs to find support on various discussion forums and communities. At times, when you are in a hurry to implement something, you may not find a vendor to help you—that is where the proprietary technologies score over OSS. With easy and immediate support from vendors, such stressful situations are avoided when one uses proprietary technologies. It would be incorrect to say that the support system for all open source technologies is not all that strong; but this is the case for some cutting-edge products that have not been widely adopted and are still in an evolving stage. Those are the technologies which make one think about whether to use open source or not. As an example, Apache has come up with Cassandra. To implement Cassandra, I will have to do a lot of R&D. I would rather have a vendor, who could implement it for me. Involvement of a vendor saves a lot of time and energy; particularly, if the project is on a tough timeline, and there is no scope to experiment.

"Other than that, we face the issue of a lack of talent. People with an understanding of open source products are very much in demand these days. It is more difficult to get someone working on Ruby on Rails or Python, for instance, than getting someone working on technologies like .NET."

But the human resource issue is managed by the company in a beautiful way. It trains people in the technologies that it works with. Sangoi states, "While hiring, we look for people who are open to learning. We also have some internal champions, who keep driving the projects that are based on open source technologies and also bring others along with them. These internal champions help in providing on-job training."

Community interaction is important

Although Sangoi and his team do not contribute to the community much, they definitely harness its power. Nilesh shares, "Once, when we began using Redis for memory based caching, we came across some roadblocks while integrating it with our back-end. We put our problems up on the community sites and we got some great help—and that too, just in an hour." END